

Xiang Zhao

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RESEARCH INTERESTS

Video-centric multimodal learning, focusing on models that interpret multimodal inputs and capture relationships among language, visual content, motion, and temporal structure for controllable video generation or action prediction.

EDUCATION

Northeastern University (China)

M.Eng. in Control Engineering

B.Eng. in Automation

Shenyang, China

2016.09 – 2019.01

2012.08 – 2016.06

PROFESSIONAL EXPERIENCE

Alibaba Hujing Digital Media & Entertainment Group

Senior Algorithm Engineer — Video & Image Generation

Beijing, China

2023.08 – 2026.05

Audio-Conditioned Mask-Free Lip Synchronization via Video Diffusion

- Built an audio-conditioned video diffusion system for mask-free lip synchronization using pseudo-paired training data derived from MEAD; used separate high- and low-noise training strategies to preserve global appearance while recovering fine lip detail.
- Applied distribution-matching distillation to reduce inference from 50 to 4 steps while retaining localized mouth control; inference requires neither masks nor mouth landmarks.

VLM-Guided Reinforcement Learning for Expression Editing

- Applied Flow-GRPO with a reward combining VLM-based expression alignment and facial-identity preservation to adapt an expression-editing model pretrained on synthetic paired renderings to real faces, without pixel-aligned real targets.
- Interpolated the parameters of the RL and SFT checkpoints to balance expression control and identity preservation, selecting the final operating point through qualitative side-by-side evaluation.

Personalized Lip Synchronization with Audio-Visual Contrastive Alignment

- Developed a personalized StyleSync-based system from ~1 minute of reference video and added contrastive alignment between audio and lip representations; reduced RTX 4090 inference time by 50% to 0.5 seconds per video-second and deployed the system in production.

Xiaoice (a Microsoft-incubated AI company)

Senior Algorithm Engineer — Computer Vision

Beijing, China

2022.06 – 2023.07

High-Fidelity Face Swapping for Virtual Idols

- Designed identity compensation and self-supervised color blending to address structural identity drift and post-fusion color mismatch; deployed the system for virtual-idol content on Douyin.

AMAP (Alibaba Group)

Senior Algorithm Engineer — Machine Learning

Beijing, China

2021.08 – 2022.05

- Combined contrastive Link2Vec representations with Transformer feature fusion for ETA prediction, improving the nationwide high-quality ETA rate by 1 percentage point versus the previous production model.

BIGO Technology

Senior Algorithm Engineer

Beijing, China

2020.05 – 2021.08

Unpaired Style Transfer for Real-Time Mobile Deployment

- Constructed pseudo-paired data through StyleGAN inversion and distilled image-to-image style transfer models for real-time mobile inference.

Multi-Person Drivable Photo Singing Sticker — Project Lead

- Led the server-side project from model training and super-resolution through acceleration, service integration, and deployment; delivered BIGO's first simultaneous multi-person sticker and reached over 100,000 daily active users in Southeast Asia during launch.

Direct Audio-to-Expression Prediction for Lip Synchronization

- Mapped audio directly to a 20-D vector of 3D facial-expression parameters, bypassing speech-to-phoneme conversion to reduce latency; deployed a GAN-based lip-synthesis system supporting profile views in TAL's AI classroom.

MASTER'S THESIS

M.Eng. Thesis, Northeastern University (China)

2018.12

LSTM-Based Power Load Forecasting for Steel Enterprises

- Developed an LSTM-based Seq2Seq model for power-load forecasting in steel enterprises, increasing forecasting accuracy on the test set from 97.3% for the LSTM baseline to 98.6%.

PUBLICATIONS & PATENTS

Publication

- [1] Xinyi He, Jin Lang, Yanyan Zhang, Gengnan Zhao, and **Xiang Zhao**. "Power Load Forecasting Method of Iron and Steel Enterprises Based on Improved Long Short-Term Memory Neural Network" (in Chinese). *Metallurgical Industry Automation*, 46(3): 48–56, 2022. DOI: 10.3969/j.issn.1000-7059.2022.03.007.

Granted Patents

- [1] **Xiang Zhao**, Yule Li, and Wei Xiang. "Model Training and Image Processing Method, Device, Equipment and Medium." Chinese Invention Patent [CN 112767287 B](#), granted December 17, 2024.
- [2] Lixin Tang, Jin Lang, and **Xiang Zhao**. "Ultra-Short-Term Wind Power Prediction Method Based on Deep Learning." Chinese Invention Patent [CN 109615146 B](#), granted February 7, 2023.

SELECTED OPEN-SOURCE PROJECTS

- **HeadSwap** ([GitHub](#)): high-fidelity head-swap implementation.
- **Simple-LoRA** ([GitHub](#)): lightweight LoRA training codebase for Stable Diffusion.
- **Triton-Torch-Custom** ([GitHub](#)): example of deploying PyTorch custom operators with Triton Inference Server.

SKILLS

- **Methods:** diffusion and flow-matching models, contrastive learning for multimodal alignment, VLM-guided post-training, and model distillation.
- **Systems & programming:** PyTorch, distributed training, Triton Inference Server deployment; Python, C/C++, SQL, MATLAB.
- **Languages:** Chinese (native), English (proficient).

HONORS & LEADERSHIP

- **Honors:** Third Prize, Fashion AI Global Challenge (Top 7%, Alibaba Cloud, 2018); Second-class Scholarship (Top 10%, Northeastern University (China), 2013).
- **Leadership:** Director, LearnHub Club, Northeastern University (China), 2018.09 – 2019.06.